Hieroglyphic EmotionNet: Sentiment and Intention Analysis of Ancient Egyptian Texts

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Abstract

This project proposes *Hieroglyphic EmotionNet*, an AI-powered system for sentiment and intention analysis of ancient Egyptian hieroglyphic texts. The system integrates **computer vision** (hieroglyph segmentation and classification), **natural language processing** (transliteration, translation, and language modeling), and **sentiment/intention modeling** using transformer-based architectures (BERT, RoBERTa, T5). The pipeline first segments hieroglyphic inscriptions into individual glyphs, classifies them, and converts them into transliterations and translations in both English and Arabic. It then analyzes the emotional polarity (positive, negative, neutral) and underlying communicative intent (e.g., offering, prayer, mourning) of the text. The resulting framework aims to provide Egyptologists, historians, and linguists with a tool for better understanding the emotional and intentional content of ancient inscriptions. Additionally, the project will contribute a labeled dataset and codebase for research in ancient language computational semantics, bridging digital humanities and AI.

1. Introduction of the Study

The ancient Egyptian language, part of the Afro-Asiatic language family, was in use from approximately 3200 BCE to 1100 CE (Allen, 2014). Its historical development spans six phases: Archaic Egyptian, Old Egyptian, Middle Egyptian, Late Egyptian, Demotic, and Coptic. Middle Egyptian (2100–1600 BCE) maintained a "classical" status for the production of historical and religious texts even after its decline as a spoken language, which is why this project focuses primarily on Middle Egyptian, supplemented with Old Egyptian for coverage.

Ancient Egyptian employed four writing systems: hieroglyphic, hieratic, demotic, and coptic. *Hieroglyphic* script consists of pictorial signs mostly carved in stone for monumental contexts, whereas *hieratic* is a cursive form used on papyri. *Demotic* was a late-stage cursive script, and *Coptic* incorporated the Greek alphabet with additional Demotic characters. This project focuses

on hieroglyphic inscriptions (or hieratic texts transcribed into hieroglyphic form) because they correspond to the language phases of interest.

Hieroglyphs fall into three categories: ideograms (depict the word they represent), phonograms (express phonetic sounds), and determinatives (provide semantic context without being pronounced) (Allen, 2014). To standardize glyph identification, we use the Gardiner code, an alphanumeric system compiled by Alan Gardiner, which categorizes each hieroglyph with a letter and number (e.g., A1, N35a).

Prior work such as the Hieroglyphic Transformer (ACL 2024) has demonstrated accurate segmentation, transliteration, and translation of hieroglyphs. However, these systems do not yet provide semantic interpretation, such as sentiment polarity or communicative intention. This gap motivates our project, which integrates modern NLP techniques to extract emotional and intentional meaning from ancient texts.

2. Problem Statement

Existing computational pipelines for hieroglyphic texts focus on visual recognition, transliteration, and literal translation. While these tools are effective for producing readable text, they fail to capture the **emotional tone or intent** expressed in inscriptions. Ancient texts often convey prayers, praises, warnings, or mourning, and understanding these subtleties is essential for historical and linguistic research. The absence of an automated system for **sentiment and intention analysis** limits the depth of digital Egyptology studies.

3. Objectives of the Study

Main Objective:

Develop an AI-based framework capable of performing **sentiment and intention analysis** on ancient Egyptian hieroglyphic texts and providing **translations in both Arabic and English**.

Specific Objectives:

1. Extend the Hieroglyphic Transformer pipeline to process hieroglyphic images into transliterations.
2. Translate hieroglyphic transliterations into **both Arabic and English** using a fine-tuned seq2seq transformer or retrieval-augmented approach.
3. Map translations to semantic categories using ancient Egyptian–English/Arabic lexicons (Thesaurus Linguae Aegyptiae).

4. Implement sentiment analysis (positive, negative, neutral) using transformer-based models (BERT, RoBERTa) for English and (AraBERT) for Arabic.
5. Perform intention detection (e.g., offering, prayer, mourning) using a multi-label classification approach (T5).
6. Create a dashboard/interface visualizing hieroglyphic text, transliteration, **dual-language translation**, sentiment, and intention.
7. Contribute an annotated dataset and trained models for future Egyptology and AI research.

4. Literature Review

Recent advancements in artificial intelligence have revived interest in the computational study of ancient languages. In the context of Egyptian hieroglyphics, deep learning has enabled new approaches for interpreting visual inscriptions, transliteration, and translation — bridging the gap between digital humanities and modern machine learning.
Recent research in **Hieroglyphic Recognition and Translation** has demonstrated the potential of neural architectures for decoding ancient scripts. The *Hieroglyphic Transformer* model [1)] introduced an end-to-end framework for transliteration and translation, showing strong contextual awareness through transformer-based modeling. Earlier works such as *Image-Based Hieroglyphic Character Recognition* [2)] established the role of convolutional neural networks (CNNs) in identifying hieroglyphic symbols from scanned inscriptions, while the *Focused Generic Segmentation and Cross-Validation Voting* approach [(3)] enhanced segmentation accuracy and robustness through ensemble learning methods.

In the area of **Sentiment and Intention Analysis**, *Emotion Recognition Using Transformer Architectures* [(4)] demonstrated the effectiveness of attention-based models for emotion classification in multilingual contexts. The *Hybrid AI-Driven Sentiment Interpretation Model* [5)] further combined BERT embeddings with lexicon-based processing for more interpretable sentiment analysis. Additionally, the *Aspect-Based Sentiment and Intention Modeling for Historical Arabic Texts (LT4HALA 2024)* study [6)] highlighted the potential of domain-specific pre-training (AraBERT, MARBERT) in improving sentiment and intention detection for linguistically rich historical corpora.
Collectively, these studies have shown the potential of deep learning for linguistic interpretation; however, they remain limited to either translation or emotion analysis tasks in modern languages. No existing work integrates **computer vision, translation, and sentiment/intention modeling** into a unified system for ancient Egyptian hieroglyphic texts. This project, **Hieroglyphic EmotionNet**, addresses this gap by combining visual, linguistic, and semantic modeling to perform end-to-end hieroglyphic understanding and emotional interpretation in both English and Arabic.

5. Research Gaps

While these works demonstrate success in glyph recognition and translation, they primarily focus on syntactic processing, not semantic interpretation.
No current system extends hieroglyphic translation to emotion, sentiment, or intention detection, and existing models focus on single-language output (mainly English).
Moreover, integration between computer vision (for glyph extraction) and language understanding (for semantic labeling) remains limited.
Our proposed system addresses these gaps by:

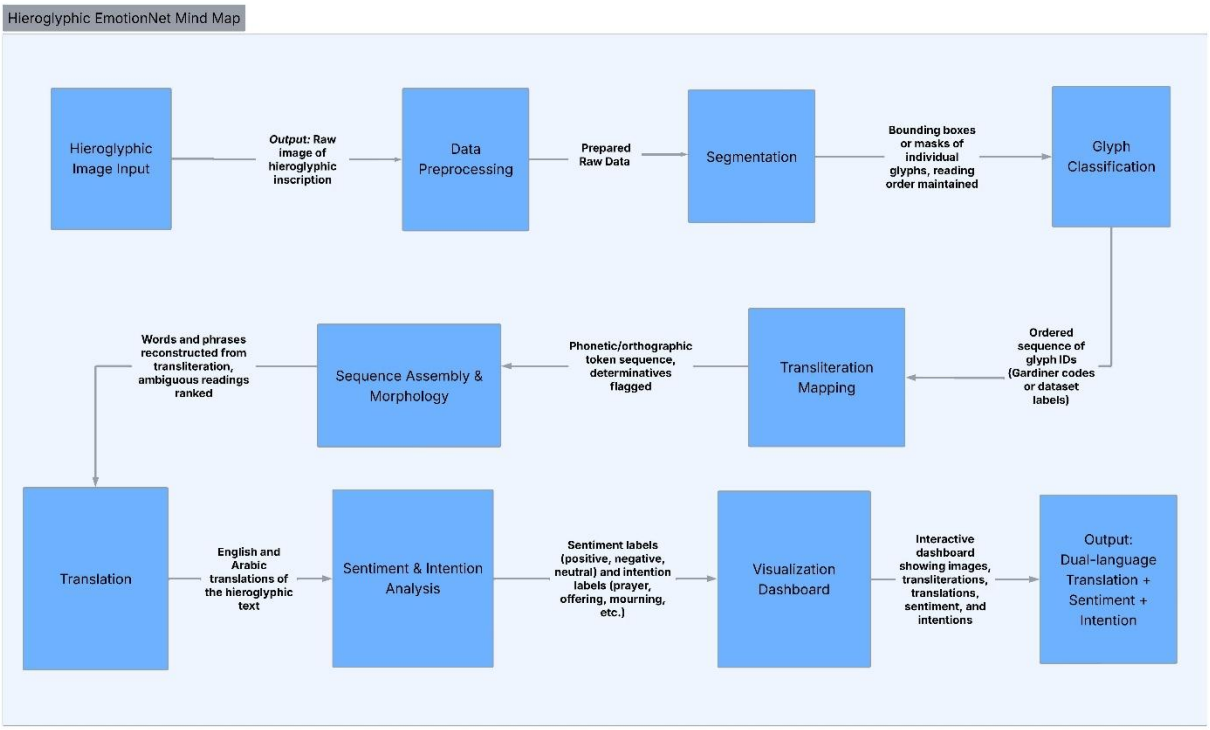
- Extending hieroglyphic translation into sentiment and intention analysis using transformer-based NLP models (RoBERTa, BERT, AraBERT).
- Producing dual-language translations (English and Arabic) for broader accessibility and cultural relevance.
- Combining CNN-based vision models (ResNet/GlyphNet) with transformer-based language models (T5, AraT5) to create a unified multimodal pipeline for end-to-end hieroglyphic understanding.

6. Proposed Methodology

The proposed methodology follows a multi-step pipeline from hieroglyphic images to sentiment and intention analysis in both English and Arabic:

1. **Image Preprocessing (CV Quality Control):**
   This initial step ensures that the raw image of the hieroglyphic inscription, which is often affected by age, damage, and poor lighting, is cleaned and optimized before the system attempts to detect any actual glyphs.
2. **Segmentation :**
   Extract individual glyphs from scanned images using OpenCV or instance segmentation models such as Mask R-CNN and Detectron2. Initial training will use clean datasets (e.g., Unas, Fuentes-Ferrer).
3. **Glyph Classification :**
   Classify each isolated glyph into its canonical identifier using CNN-based models (ResNet, Xception) or lightweight networks (GlyphNet). Transfer learning will be employed if datasets are small
4. **Transliteration Mapping :**
   Map glyph IDs to phonetic or orthographic tokens using the Gardiner sign list and JSesh transliteration standards. Determinatives will be identified and handled separately.
5. **Sequence Assembly & Morphological Processing :**
   Assemble transliterations into words and phrases using lexicon lookups (Thesaurus Linguae Aegyptiae) and language models (n-gram or neural LMs) for disambiguation.
6. **Translation :**
   Convert transliterations into **English and Arabic translations** using rule-based lexicons and transformer-based seq2seq models (T5, MarianMT, BART, AraT5). Retrieval-augmented generation may be used for higher accuracy.

7. **Sentiment & Intention Analysis :** Perform sentiment classification (positive, negative, neutral) and detect intentions (e.g., offering, prayer, mourning) using fine-tuned transformer models: **ROBERTa/BERT** for English and **AraBERT** for Arabic, with **T5-based multi-label classification** for intention.
8. **The Intention Revealing Step:**
   Immediately following the Sentiment Classification, the pipeline executes the **Intention Classification** task. The model, specifically a **T5-based multi-label classifier**, analyzes the English and Arabic translations to categorize the text's underlying communicative purpose (e.g., **offering, prayer, mourning, declarative/informative**). This final classification is crucial as it leverages the domain-specific labels from the Custom Annotated Corpus, providing a deeper, context-aware interpretation of the text beyond simple emotional polarity.



7. Datasets

1. [Unas Hieroglyph Dataset] — A public dataset containing labeled images of hieroglyphic symbols used for segmentation and classification model training.

2. [Fuentes-Ferrer Hieroglyphic Dataset] — A benchmark dataset featuring segmented hieroglyphs annotated with Gardiner codes, supporting glyph classification and transliteration mapping.

3. **Linguistic and Semantic Corpora:** Transformer models are fine-tuned using a multi-stage approach, incorporating both high-quality generic corpora and a domain-specific resource:

- **Arabic Sentiment Adaptation:** The AraBERT model is fine-tuned for robust general Arabic sentiment using powerful external corpora, such as the ASTD (Arabic Sentiment Tweets Dataset) and BRAD (Books Reviews in Arabic Dataset).
- **English Sentiment Adaptation:** English transformer models like ROBERTa/BERT are initialized and adapted using broad corpora, including the IMDb Movie Review Dataset and SST-2 (Stanford Sentiment Treebank).
- **Domain Adaptation (Crucial Step):** The final layer of adaptation for both sentiment and intention is achieved using the Project's Custom Annotated Corpus, which provides the unique, manually labeled sentiment and intention context specific to ancient Egyptian texts.
- Example for corpus structure:

Hieroglyphic Source ID	**English Translation**	**Arabic Translation**	**Sentiment Label**	**Intention Label**
TLA-12345	"May the King give an offering..."	" يأمر الملك بتقديم قربان..."	**Positive**	**Offering**

These datasets collectively ensure the robustness of both the visual and linguistic components of the pipeline and enable supervised fine-tuning for translation and sentiment analysis.

8. Expected Outcomes

- Accurate segmentation and classification of hieroglyphic glyphs.
- Transliteration of hieroglyphic inscriptions into machine-readable sequences.
- High-quality translations of inscriptions into **English and Arabic**.
- Sentiment analysis indicating emotional tone (positive, negative, neutral) for each inscription.
- Intention classification highlighting underlying communicative purposes (e.g., offering, prayer, mourning).
- A user-friendly visualization dashboard for scholars and researchers.
- A curated dataset of transliterations, translations, and sentiment/intention labels for future research.

9. Tools & Technologies

The implementation of *Hieroglyphic EmotionNet* combines advanced computer vision, natural language processing, and deep learning frameworks to deliver an integrated end-to-end system.

- **Computer Vision & Segmentation:** Hieroglyph segmentation and extraction will utilize *OpenCV*, *Detectron2*, and *Mask R-CNN* frameworks built with *PyTorch* and *TensorFlow*, ensuring precise identification of individual glyphs within images.
- **Glyph Classification:** Convolutional neural networks, including *ResNet*, *Xception*, and the specialized *GlyphNet* architecture, will be trained through transfer learning to classify glyphs into standardized Gardiner codes.
- **Transliteration & Lexical Mapping:** The *JSesh* transliteration standard and *Thesaurus Linguae Aegyptiae (TLA)* lexicon will be used to map classified glyphs to their phonetic and semantic representations.
- **Translation:** Transformer-based sequence-to-sequence models such as *T5*, *BART*, and *MarianMT* will be fine-tuned for English translation, while *AraT5* will generate parallel Arabic translations.
- **Sentiment and Intention Analysis:** Fine-tuned transformer models (*RoBERTa*, *BERT*, and *AraBERT*) will analyze emotional polarity (positive, negative, neutral), while a *T5* multi-label classifier will detect communicative intentions such as offering, prayer, or mourning.
- **Visualization:** A user-friendly dashboard built using *Streamlit* and *React* will visualize the full pipeline results—displaying the original hieroglyphic image, segmentation results, transliteration, dual-language translations, and semantic annotations.

10. Expected Challenges

- **Determinatives & Logograms:** Handling non-pronounced signs and symbols representing whole words.
- **Missing Vowels:** Ancient Egyptian script lacks vowels, increasing ambiguity in transliteration.
- **Data Scarcity:** Limited parallel corpora for transliteration → translation in both English and Arabic.
- **Translation Accuracy:** Maintaining semantic and syntactic fidelity in both languages.
- **Cultural Context:** Ancient texts may express sentiment differently than modern language models interpret.
- **Model Generalization:** Ensuring models trained on clean datasets can handle damaged or noisy inscriptions.
- **Multi-language Sentiment Analysis:** Adapting English and Arabic sentiment models to domain-specific texts.

11. Project Schedule & Milestones

Phase	Duration	Key Milestones (Deliverables)	Technologies
Phase I: Data & Vision Pipeline	8 Weeks	Glyph segmentation model trained and validated (Step A). Glyph classification model trained and tested on target datasets (Step B). Initial transliteration mapping framework established (Step C).	Detectron2, Mask R-CNN, ResNet, Xception, OpenCV, JSesh
Phase II: Language & Translation Core	8 Weeks	Full sequence assembly and morphological processing implemented (Step D). Dual-language (English/Arabic) translation models fine-tuned and evaluated (Step E). Parallel corpus dataset created and labeled for subsequent phases.	T5, MarianMT, BART, AraT5, Thesaurus Linguae Aegyptiae
Phase III: Semantic Modeling	6 Weeks	Sentiment Analysis models (ROBERTa/BERT/AraBERT) fine-tuned for ancient text domain (Step F). Intention Classification model (T5 multi-label) trained and validated (Step F). Comprehensive sentiment and intention annotated dataset finalized.	BERT, ROBERTa, AraBERT, T5 Multi-label
Phase IV: Integration & Deployment	4 Weeks	End-to-end pipeline integrated, tested, and optimized for speed and accuracy. User-friendly visualization dashboard developed (Streamlit/React) (Step G). Final report, codebase, and annotated dataset released for public research[11].	Streamlit, React, Flask, PyTorch/TensorFlow

12. Scholarly and Cultural Impact

The Hieroglyphic EmotionNet is more than a technical achievement; it represents a significant advancement for the fields of Egyptology and AI.

- **Quantifying History**: By automatically classifying inscriptions by communicative intent (e.g., prayer, offering, mourning), this tool will enable historians and archaeologists to quantify changes in religious, political, and social sentiment over different chronological periods for the first time.
- **Bridging Languages**: Providing high-quality, dual-language translation in both English and Arabic significantly enhances accessibility, allowing a broader international community of scholars, students, and the general public to engage directly with ancient Egyptian content.
- **Foundational Research**: The project's most lasting contribution will be the annotated dataset and the trained transformer models. This open-source resource will serve as the first comprehensive benchmark for computational semantics in ancient languages, laying the groundwork for future AI research in digital humanities

13. References

1. Zanettin, G., *Hieroglyphic Transformer: A Deep Learning Model for Transliteration and Translation of Egyptian Hieroglyphs*, ACL Anthology, 2024.
2. El-Sawy, A., El-Bakry, H. & Loey, M., *Image-Based Hieroglyphic Character Recognition*, ResearchGate, 2019.
3. Fuentes-Ferrer, M. et al., *Recognition of Egyptian Hieroglyphic Texts through Focused Generic Segmentation and Cross-Validation Voting*, ResearchGate, 2024.
4. Akhtar, S., *Emotion Recognition Using Transformer Architectures*, *Electronics (MDPI)*, 13(9), 2024, pp. 1753–1766.
5. Ahmed, F. & Khan, N., *Hybrid AI-Driven Sentiment Interpretation Model*, SSRN Electronic Journal, 2024.
6. Dejaeghere, W. et al., *LT4HALA-1.16: Aspect-Based Sentiment and Intention Modeling for Historical Arabic Texts*, ACL Anthology, 2024.

Example Walkthrough

Consider the hieroglyphic phrase:

Step A – Segmentation (Glyph Detection):
The system detects seven individual glyphs in reading order using OpenCV / Detectron2.

Step B – Glyph Classification (Gardiner Codes):

Glyph	Predicted ID
	D21
	Z1
	N35
	M17
	G43
	M17
	M17

Step C – Transliteration Mapping:

- D21 → r
- Z1 → (ignored / determinative)
- N35 → n
- M17 → i
- G43 → w
- M17 M17 → i.i

→ **Transliteration**: rn iwii

Step D – Sequence Assembly & Morphology:

- **Lexicon lookup (TLA):** rn = "name", iwii = "comes"
- **Assembled phrase:** "The name comes forth."

Step E – Translation to Modern Language:

- **English:** "The name comes forth."
- **Arabic:** "الاسم يأتي."

Step F – Sentiment & Intention Analysis:

- **Sentiment (RoBERTa / AraBERT):** Neutral
- **Intention (T5 multi-label):** Declarative / informative

Step G – Visualization:

- Displayed in the dashboard with glyph images, transliteration, English & Arabic translation, sentiment, and intention labels.